RAJALAKSHMI ENGINEERING COLLEGE

Lavanya V 250701386-2025-CSE

L2

Started on	Monday, 31 August 2026, 9:38 PM
State	Finished
Completed on	Monday, 31 August 2026, 9:41 PM
Time taken	2 mins 42 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n, x;
    scanf("%d", &n);
    int a[n];
    for (int i = 0; i < n; i++) scanf("%d", &a[i]);
    scanf("%d", &x);

    int l = 0, h = n - 1, ans = -1;
    while (l <= h) {
        int m = (l + h) / 2;
        if (a[m] <= x) {
            ans = a[m];
            l = m + 1;
        } else {
            h = m - 1;
        }
    }
    printf("%d\n", ans);
    return 0;
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔

	Input	Expected	Got	
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course